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Carbon Services

ADM CCS1 Injection Well
Mechanical Integrity Report
March 2019
Pulsed Neutron eXtreme

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1. Pulsed Neutron eXtreme Mechanical Integrity Log

A Pulsed Neutron eXtreme (PNX) mechanical integrity inspection log was run to evaluate well integrity and formation carbon dioxide (CO_2) saturation in CCS1 on 6 March 2019. PNX is a pulsed neutron capture tool that is capable of making several different measurements sensitive to CO_2 in the formation and in the casing-formation annulus. Therefore, the PNX can be used to monitor the formation fluids as well as identify mechanical integrity problems that may allow the CO_2 to migrate up the casing annuli. The PNX was run in the Gas-Sigma-Hydrogen Index (GSH) mode from 7,124 to 200 feet (ft) in CCS1. An evaluation incorporating the PNX data with the openhole logging was made for the current formation fluids and PNX data inspected for any annulus CO_2 and/or gas. For mechanical integrity monitoring, the PNX is most effective when used as a time-lapse measurement comparing a base pass to subsequent monitoring passes. Baseline Sigma, prior to CO_2 injection, was provided using the Reservoir Saturation Tool (RST) on March and September 2011. Subsequent RST monitor passes were March, July and November 2012, July 2013 and December 2014. PNX is the latest generation pulsed neutron tool and was logged in addition to the RST on 20 July 2014. PNX monitoring passes were acquired in the CCS1 well on 25 October 2016, 8 May 2018 and the most recent 6 March 2019.

2. Analysis of PNX Data and Interpretation

The March 2019 mechanical integrity monitoring for CCS1 utilized the PNX. The PNX is the latest generation pulsed neutron tool and has been optimized for gas, sigma, and neutron porosity measurements in terms of both accuracy and precision. The PNX introduces a new fast neutron measurement, fast neutron cross section (FNXS) that is highly sensitive to CO_2 and gas volume while insensitive to water salinity and volume. A special pulse sequence used with the neutron generator allows borehole effects to be determined allowing formation capture cross section (SIGM) and neutron porosity (TPHI) to be self-compensated. Therefore, monitoring passes can be meaningfully compared, even with changing borehole fluids and any change to the completion tubulars.

CCS1 was drilled to a depth of 7,235 ft. and completed as follows:

- Bit size of 26 inches (in.) to a depth of 355 ft., 17.5 in. to 5,339 ft., and 12.25 in to a total depth of 7,235 ft.
- Intermediate casing was 13.375 in., J-55 59.5 & 66.2 pounds per foot (lb/ft) from surface to 5,339 ft.
- The final section casing is 9.625 in. from surface to total depth with N80 40 lb/ft from surface to 5,272 ft. and 13Cr80 47 lb/ft from 5,272 ft. to 7,219 ft.
- The injection tubing string is 4.5 in. 12.6 lb/ft from surface to 6,414 ft with the packer set at 6,365 ft.

Technical Background

Pulsed neutron Sigma (∑) is the thermal neutron capture cross section or the rate at which thermal neutrons are captured by the formation matrix and fluids. PNX has a pulsed neutron generator that creates high-energy neutrons. These neutrons are slowed to a thermal velocity through elastic and inelastic collisions with the nuclei of the environment's elements, at which point they can be captured. Hydrogen, with the most similar nuclei mass to the neutron, is the most effective element to slow the high energy neutrons and chlorine is very good at capturing the thermal neutrons. Therefore, water has a capture cross section, which depends largely on its salinity. Fresh water has a Sigma response of approximately 22 capture-units (cu) and salt saturated water over 130 cu. CO_2 has a very low ability to slow or capture neutrons and the Sigma response for CO_2 is near zero. Using the

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contrast in Sigma for water and CO_2 the formation water saturation (S_w) can be calculated from the Sigma measurement. The Sigma measurement is corrected for the borehole environment and therefore is only sensitive to formation fluid changes.

SIBH is the rate at which thermal neutrons are captured as measured with the earlier timing gates and shorter source-detector spacing making it primarily sensitive to the borehole, and therefore borehole and annular CO_2. CO_2 in the wellbore or well annuli will decrease SIBH.

The PNX neutron porosity measurement (TPHI) is an inverse relationship of the slowing down length of high energy neutrons emitted by the tool and responds primarily to the amount of hydrogen within the formation. Hydrogen is generally related to the fluids in the porosity (e.g., either water or oil). The carbon and oxygen nuclei of CO_2 have a much higher mass than a neutron and interactions with high energy neutrons are primarily elastic with the neutron losing very little energy similar to other formation elements. As CO_2 saturation increases replacing formation water, the hydrogen density decreases causing the neutron porosity measurement to be lower than the true porosity. The PNX neutron porosity has a depth of investigation around 10 to 12 inches. The PNX TPHI algorithm uses count ratios from the three detectors and several timing gates to correct for the borehole environment. However, CO_2 in the well annuli requires a large correction. In a large borehole, the algorithm may not be able to fully correct for the CO_2. Therefore, TPHI is primarily sensitive to CO_2 in the formation, lowering the measured neutron porosity. And in large boreholes filled with CO_2, the TPHI measurement may not fully correct for the environmental, effecting the measurements accuracy.

Gas ratio (GRAT) and Fast Neutron Cross Section (FNXS) are new measurements unique to the PNX tool and based on the measurement of gamma rays resulting from inelastic neutron interactions, which are very sensitive to CO_2 and gas. Since inelastic gamma rays can only be induced by neutrons with energies at MeV levels, they respond to formation properties that are independent of neutron porosity and Sigma. The PNX neutron generator (minitron) burst contains a high flux of high energy neutrons with any source variations measured and normalized using a compact neutron monitor (CNM) detector placed near the minitron. The gamma rays resulting from the high energy neutron interactions during the minitron burst contain inelastic, epithermal, and thermal capture events. By measuring the gamma ray events during and following a burst, the capture background can be determined and removed from the total measured gamma rays.

GRAT is defined as the capture background corrected burst gamma ray count rate normalized by the neutron monitor count rate. GRAT will increase in the presence CO_2 and/or gas either in the formation, annulus, or in the wellbore. Characterizing GRAT for various formation conditions was done to obtain the new measurement called fast neutron elastic scattering cross-section, FNXS. GRAT and FNXS are independent to SIGM and TPHI, and highly sensitive to CO_2 with very little sensitivity to water salinity or formation properties. FNXS varies linearly, decreasing in the presence of CO_2 and/or gas and can be used as an independent input to calculate the saturation of CO_2 and/or gas. David Rose et al. (2015) describes the PNX tool measurements in detail in their paper titled "*An Innovative Slim Pulsed Neutron Logging Tool*".

GRAT and FNXS are shallow reading measurements – two to four inches. The PNX log is centralized in the CCS1 well by the completion to a depth of 6,414 ft. This makes GRAT and FNXS measurements primarily sensitive to the borehole and annular space in this well with less sensitivity to the formation. In CCS1 above 6,414 ft. GRAT and FNXS cannot be used as an input to formation fluid analysis but remain very valuable well integrity measurements. Below 6,414 ft., the GRAT and FNXS measurements will be sensitive to the formation CO_2 in addition to the annular space. Where there is CO_2 in the borehole below 6,414 ft. effecting the accuracy of TPHI, FNXS can be used as an input for formation CO_2 saturation evaluation.

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Results

PNX Time-lapse Inspection Log

Figures 1, 2, and 3 show the PNX time-lapse inspection log for different sections of the CCS1 well. A brief description of the log data in each track is below.

Track 1 Formation Tops

Track 2 Measured Depth

Track 3 PNX Gamma Ray (GR), Openhole Spectral Gamma Ray (HSGR and HCGR), Bit Size (BS), Openhole Caliper (HCAL)

Track 4 Well Temperature (WTEP) and Well Fluid Density (MWFD) measured on PNX cartridge. MWFD is calculated from differential of pressure measurement at two tool depths.

Track 5 ELAN Volumetric Openhole Interpretation

Track 6 PNX Sigma Borehole (SIBH). Sigma Borehole is primarily sensitive to the near well bore environment and borehole fluids. In the presence of CO_2 in the borehole or annulus, SIBH will decrease from the water baseline. SIBH will also be sensitive to the borehole or annulus water salinity.

Track 7 Formation Sigma (SIGM). Formation Sigma has been corrected for the near well bore environment and is primarily sensitive to the formation fluids. CO_2 in the formation will decrease SIGM from the water baseline. An increase in SIGM indicates the formation fluid has a higher cross-capture section, generally higher salinity fluid.

Track 8 PNX Thermal Neutron Porosity Hydrogen Index (TPHI). PNX Neutron Porosity is sensitive to CO_2 in the formation and borehole environment. TPHI will decrease from the water baseline with the presence of CO_2 in the formation or borehole annulus. TPHI is less sensitive to formation water salinity changes than CO_2, and is used in conjunction SIGM to compute formation CO_2 when formation water salinity is changing.

Track 9 PNX Fast Neutron Cross Section (FNXS). FNXS is primarily sensitive to gas and/or CO_2 without sensitivity to water salinity and will decrease when CO_2 is present. FNXS is normalized for the well completion which corrects for borehole size, well tubing, and the casing completion. Well completion hardware such as packers have not been characterized and will be reflected in the FNXS data.

Track 10 PNX formation CO_2 saturation interpretation.

Track 11 Measured Depth

Track 12 PNX interpreted formation fluid volumes for the July 20, 2014 monitoring data. Light blue is the formation water, dark blue is a lower salinity water representing invaded fluids from the drilling mud, and red is the CO_2.

Track 13 PNX interpreted formation fluid volumes for the October 25, 2016 monitoring data. Light blue is the formation water, dark blue is a lower salinity water representing invaded fluids from the drilling mud, and red is the CO_2.

Track 14 PNX interpreted formation fluid volumes for the May 8, 2018 monitoring data. Light blue is the formation water, dark blue is a lower salinity water representing invaded fluids from the drilling mud, and red is the CO_2.

Track 15 PNX interpreted formation fluid volumes for the March 6, 2019 monitoring data. Light blue is the formation water, dark blue is a lower salinity water representing invaded fluids from the drilling mud, and red is the CO_2.

Track 16 Well Schematic showing tubulars, packers and perforations.

Track 17 Formation Tops

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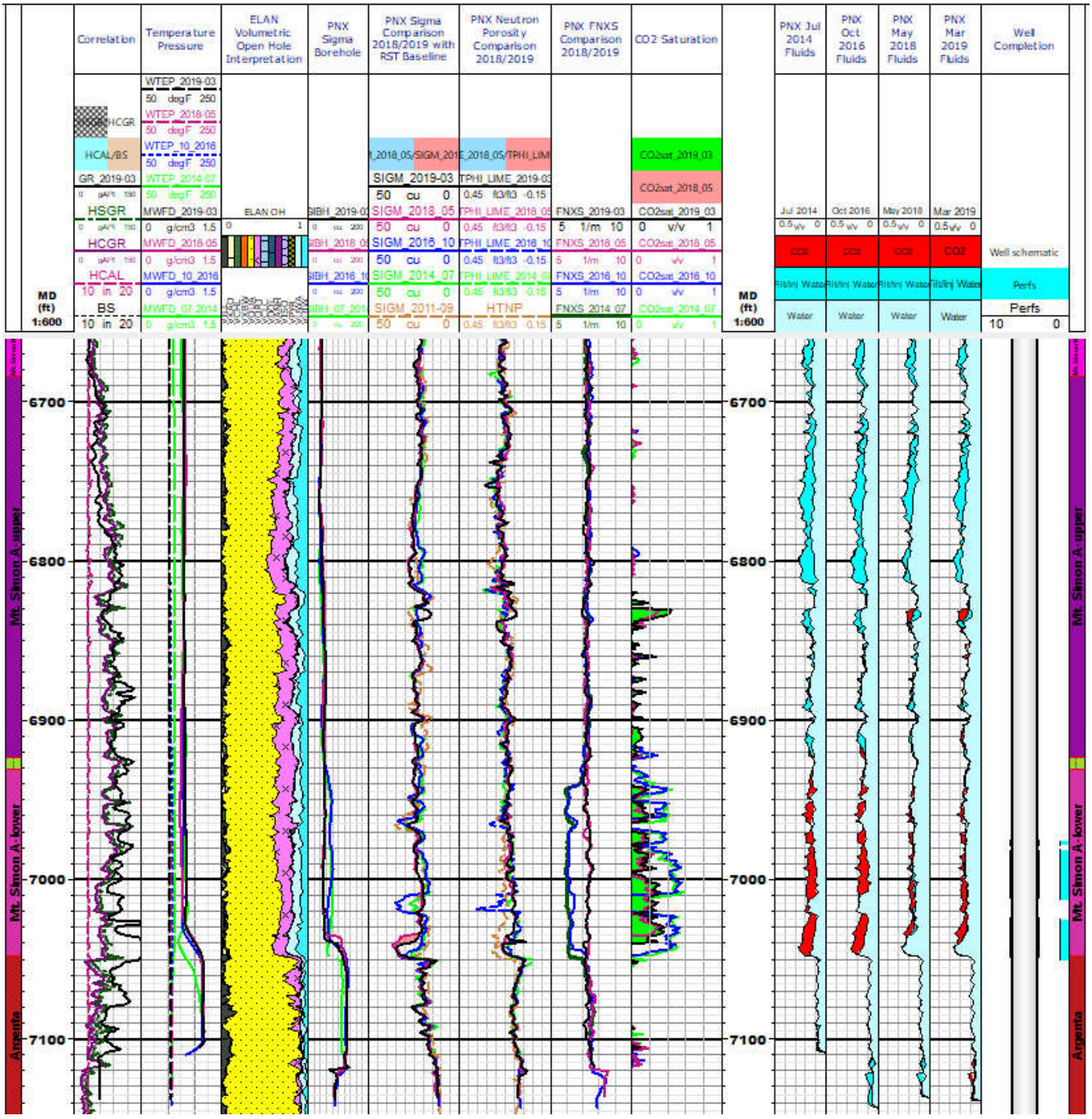


Figure 1: PNX time-lapse inspection log with outputs sensitive to CO_2 and/or gas for the depth range 6,660 to 7,124 ft.

The Illinois Basin – Decatur Project has injected approximately one million tonnes of CO_2 into the lower Mt. Simon through perforations in the CCS1 well over a three-year period from November 17, 2011 to November 26, 2014. The CCS1 well was drilled and completed in 2009. The completion diagram on the right of figures 1, 2 and 3 show the completion tubing, casing and perforated injection zones in the lower Mt. Simon A. The wellbore fluid remains CO_2 after injection to the bottom of the perforated

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interval and 1.16 g/cm^3 brine below, presumably with dissolved CO_2. The post injection water – CO2 interface below the perforations moved from 7,049 ft. during injection to around 7,033 ft in October 2016 and 7,026 ft in May 2018 and March 2019.

Inspection of the PNX time-lapse measurements sensitive to borehole and annular space are used to evaluate well integrity and detect CO_2 in the annular space or formations outside of the injection zone. Some of the measurements sensitive to the annular space are Sigma borehole (SIBH), and Fast Neutron elastic scattering Cross-Section (FNXS), and Thermal Neutron Porosity (TPHI), all decreasing with the presence of CO_2.

TPHI and FNXS measurements are corrected for the borehole environment. In the single casing portion of the well below the tubing shoe at 6,414 ft., TPHI and FNXS are primarily sensitive to formation CO_2 and can be partially corrected for the annular space. SIBH is more sensitive to the borehole environment and has a correction for the formation. In order to detect possible annular CO_2, other raw or uncorrected measurements like Sigma Short Spacing Near Apparent (SSNA) and capture background corrected burst gamma ray (GRAT) are also inspected.

Figure 1 shows the PNX time-lapse inspection log for the depth range 6,660 to 7,124 ft and Figure 2 for the depth range 6,300 to 6,725 ft. Time-lapse PNX measurements, SIBH, FNXS and TPHI in July 2014 show CO_2 in the borehole to the bottom of the perforated interval at 7,049 ft. with a transition to water around 7,083 ft. CO_2 injection stopped in November 2014 and the October 2016 time-lapse PNX measurements show the CO_2 - water contact had move up to around 7,033 ft. and to 7,026 ft in May 2018 and March 2019 with a shorter transition zone to borehole water below the perforated interval. Some CO_2 is detected in the interval 6,828 to 6,830 ft. and indication of possible CO_2 up to 6,648 ft. There is no detectable CO_2 above 6,648 ft. to the tubing shoe at 6,414 ft.

The PNX is centered in the wellbore by the tubing above 6,414 ft. SIBH remains sensitive to the borehole and annular spaces. FNXS is more sensitive to the annular spaces and less to the formation due to its shallow depth of investigation. TPHI is primarily sensitive to the formation up to 5,339 ft. Above 5,339 ft. the wellbore bit size was 17.5 in. and TPHI loses much of its sensitivity to the formation. Inspection of the PNX time-lapse measurements SIBH, TPHI and FNXS show no detected CO_2 in the wellbore annulus in the tubing section of CCS1.

Figure 3 shows the PNX time-lapse inspection log for the depth range 4,560 to 6,300 ft. The time-lapse measurements SIBH, FNXS and TPHI repeat very well for the Oct 2016, May 2018 and March 2019 logs. There are some differences in the July 2014 TPHI and FNXS measurements particularly in the 17.5 in. borehole above 5,339 ft. The 2014 tool was a pre-commercial proto-type which was logged in addition to the RST as part of the tool's development. This proto-type tool failed at 3,000 ft and began having intermittent issues as deep as 6,000 ft. Review of all the near wellbore measurements shows no detectable CO_2 in the annular space in the tubing section of the well.

There are four depths with large increases in FNXS indicating a change or additional hardware in the completion. These depths are 4,461, 5,588, 6,140 and 6,326 ft. The increase at 6,326 indicates the position of the Pressure Temperature Gauge Mandrel and the other three are the positions of the monitoring Geophones.

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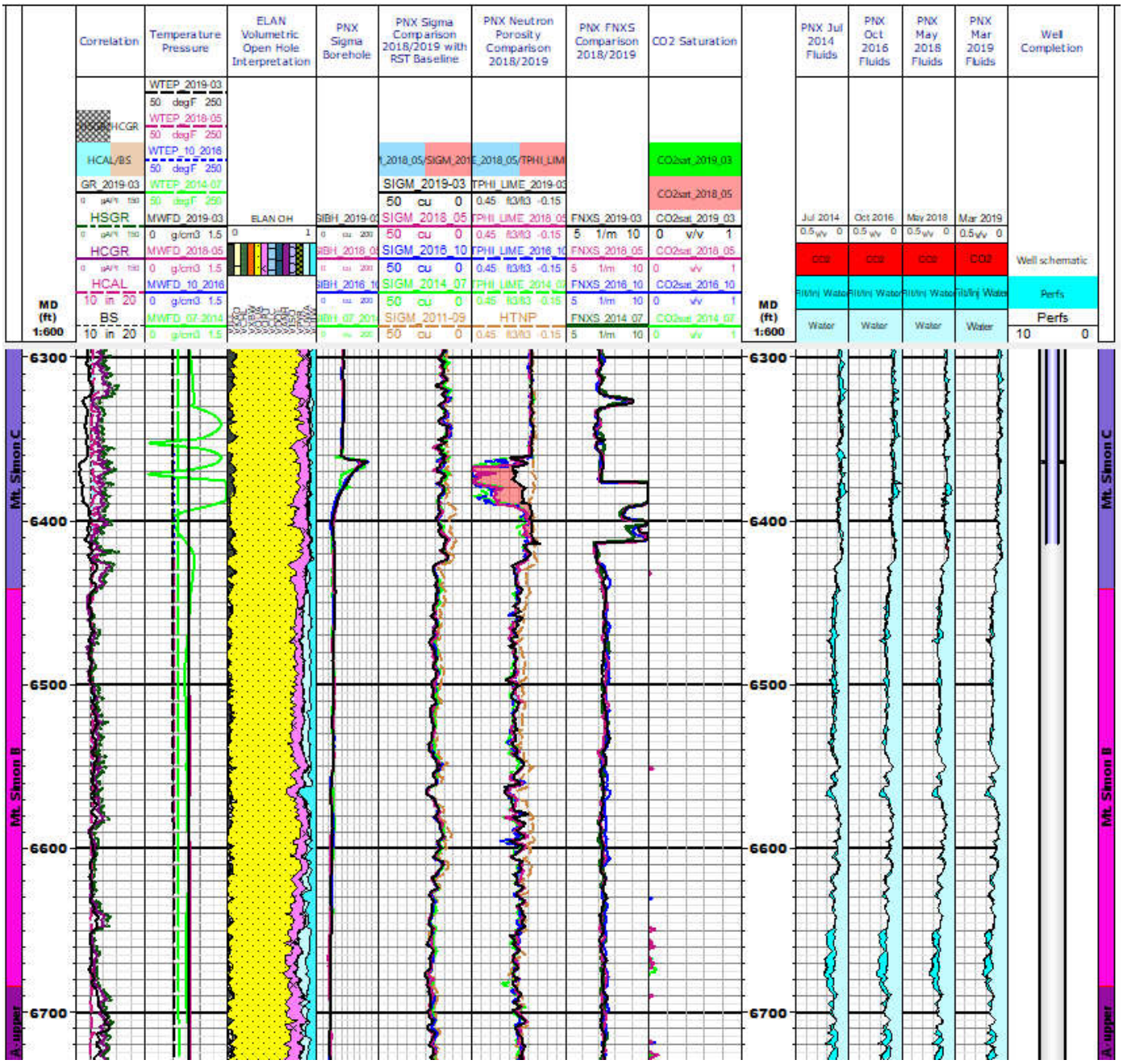


Figure 2: PNX time-lapse inspection log with outputs sensitive to CO_2 and/or gas for the depth range 6,300 to 6,725 ft.

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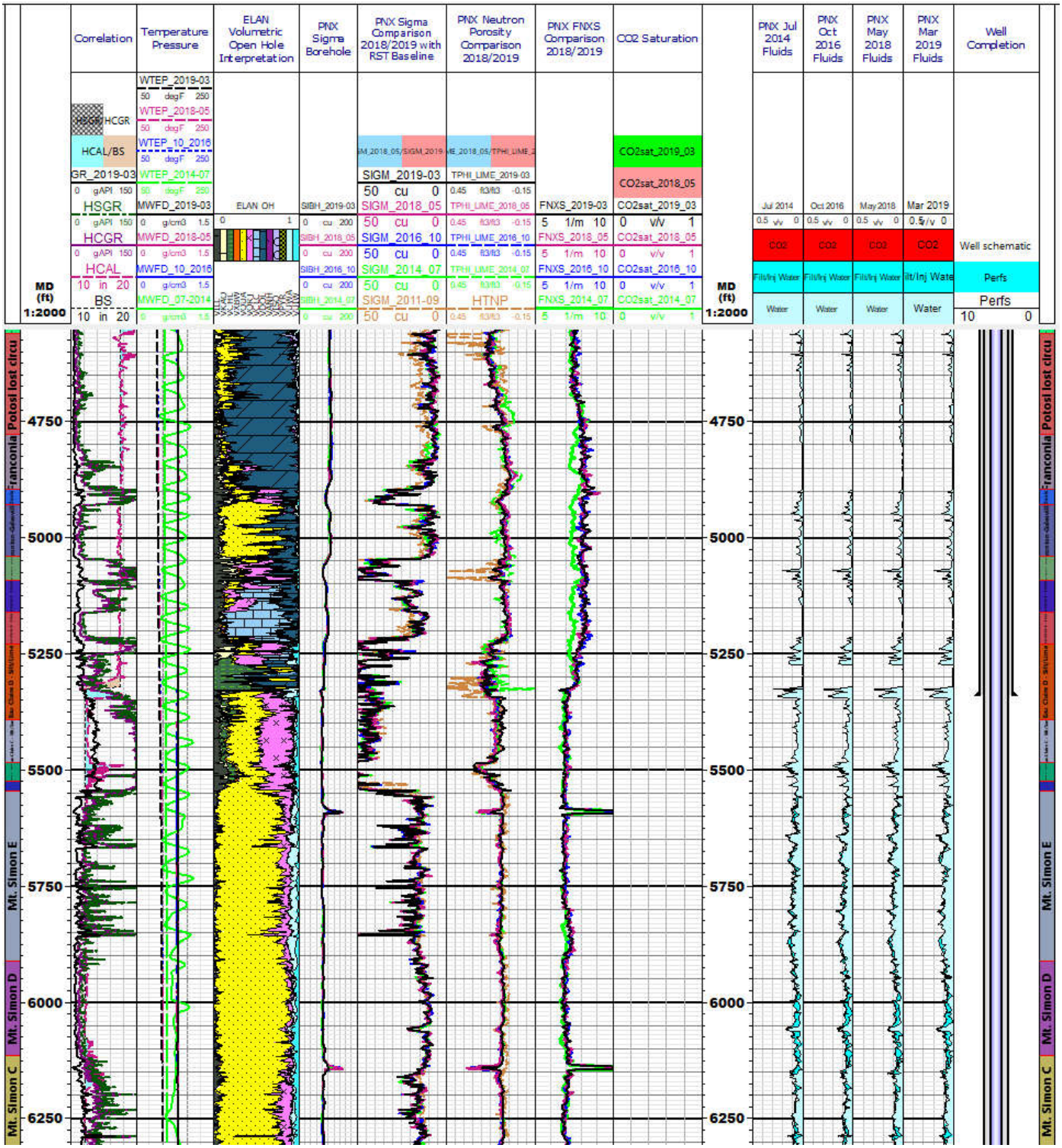


Figure 3: PNX time-lapse inspection log with outputs sensitive to CO_2 and/or gas for the depth range 4,560 to 6,300 ft.

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PNX Formation Fluid Analysis

Figures 4 and 5 show the formation fluids evaluation incorporating the PNX or RST* reservoir saturation tool data with the openhole logging evaluation. The baseline monitoring logs were measured pre-CO_2 injection in March 2011 and September 2011. Eight subsequent surveys were made with the most recent May 2018. Monitoring passes logged with the RST were Mar. 2011, Sep. 2011, Mar. 2012, Jul. 2012, Nov. 2012, Jul. 2013, Jul. 2014, and Dec. 2014. Both PNX and RST were run Jul. 2014 and PNX surveys were logged Oct. 2016, May 2018 and March 2019. A brief description of the log data in each track of the formation fluid analysis is below.

Track 1	Formation Tops
Track 2	Measured Depth
Track 3	PNX Gamma Ray (GR), Openhole Spectral Gamma Ray (HSGR and HCGR), Bit Size (BS), Openhole Caliper (HCAL).
Track 4	Well Temperature (WTEP) and Well Fluid Density (MWFD) measured on PNX cartridge. MWFD is calculated from differential of pressure measurement at two tool depths
Track 5	Formation Sigma (SIGM) all RST and PNX.
Track 6	PNX Formation Sigma (SIGM)
Track 7	PNX Neutron Porosity (TPHI).
Track 8	PNX Fast Neutron Cross Section (FNXS). FNXS is primarily sensitive to gas and/or CO_2 without sensitivity to water salinity and will decrease when CO_2 is present. FNXS is normalized for the well completion which corrects for borehole size, well tubing, and the casing completion. Well completion hardware such as packers have not been characterized and will be reflected in the FNXS data
Track 9	ELAN Volumetric Openhole Interpretation.
Track 10	CO_2 Effective Porosity Saturation. The CO_2 saturation is calculated from the formation fluid evaluation.
Track 11 - 22	Formation Fluid Analysis for the Effective Porosity using RST or PNX monitoring data. CO_2 volume is shaded red, Formation water is light blue and fresher Drilling Mud Filtrate shown in dark blue.
Track 23	Well Schematic showing tubulars, packers and perforations.
Track 24	Formation Tops

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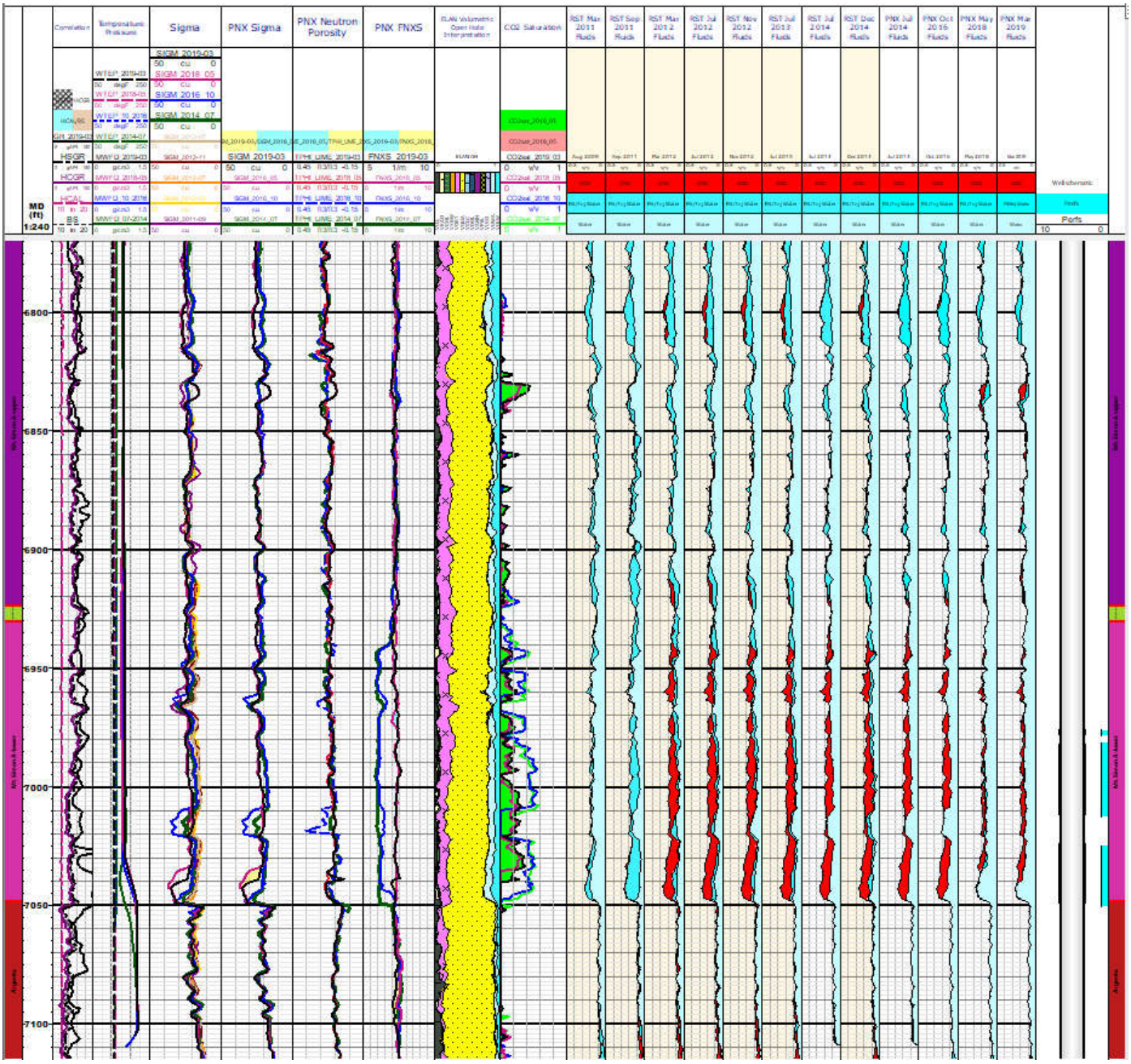


Figure 4: RST and PNX time-lapse fluids interpretation for the depth range 6,775 to 7,085 ft.

Figure 4 shows the RST and PNX time-lapse fluids interpretation for the depth range 6,760 to 7,085 ft. from the baseline log in March 2011 to March 2019. CO_2 injection began November 17, 2011 in the lower Mt. Simon Formation through perforations at 6,976 to 6,978 ft, 6,982 to 7,012 ft, and 7,024 to 7,049 ft. CO_2 saturation quickly increased near the injection zone and remained constant throughout the injection period. CO_2 was largely contained in the following zones during injection:

- 6,939 – 6,947 ft. with average CO_2 saturation of 34%
- 6,951 – 6,964 ft. with average CO_2 saturation of 43%

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- 6,969 – 6,977 ft. with average CO_2 saturation of 37%
- 6,980 – 7,011 ft. with average CO_2 saturation of 52%
- 7,020 – 7,047 ft. with average CO_2 saturation of 50%

CO_2 saturations around the injection zone remained mostly constant during the nearly two-year period post CO_2 injection, November 2014 to October 2016. There are three main differences observed in the October 2016 log. 1) There are some free-phase CO_2 buoyancy effects with the bottom depth of CO_2 saturation for all zones listed above moving upwards a few feet above apart from 6,969 – 6,977 ft. with no change. 2) A small amount of CO_2 is observed in Oct 2016 above the injection zones up to 6,915 ft. 3) The October 2016 log shows an influx of formation water between the lower two perforations in the interval 7,021 to 7,006 ft indicated by the increase in the SIGM, TPHI and FNXS measurements in this zone into the near wellbore.

The May 2018 interpretation showed a significant drop in the CO_2 saturation in the Lower Mt Simon around the injection interval from the October 2016 log. The maximum CO_2 saturation in the interval 6,940 – 7,050 ft was 56% in October 2016 and had dropped to 31% by May 2018. The CO_2 saturation average was 29% in October 2016 and had dropped to 11% by May 2018. These CO_2 saturations are similar to those observed in the VW1 observation well during the injection period. High CO_2 saturations concentrated near the injection well may have remained until CO_2 saturations further from the well had time to diffuse to a more uniform level in the Lower Mt Simon. The March 2019 PNX survey showed the CO_2 saturation near the injection interval to remain about the same as the May 2018 survey. The main difference being the interval increased in height. The May 2018 interpretation shows the majority of the CO_2 in the interval 6,940 – 7,035 ft and a small amount at 6,920 – 6,924 ft. The March 2019 interpretation shows the lower interval expanded 5 ft. to 6,940 – 7,040 ft and the upper interval expanded 16 ft. to 6,904 – 6,924 ft.

The May 2018 PNX log shows a new interval in the Upper Mt Simon A, 6,840 to 6,830 ft. with CO_2. This interval had a maximum CO_2 saturation of 30% and average of 19%. The nearby IL-ICCS project is injecting CO_2 into the Upper Mt. Simon A and Mt Simon B formations, so this new CO_2 is likely a result of the IL-ICCS CO_2 storage. The IL-ICCS project began CO_2 injection in the Upper Mt Simon A and Mt Simon B formation on April 7, 2017 and had injected 687,440 tonnes at the time of the May 2018 CCS1 log. The March 2019 PNX log shows an increase in CO_2 in this same Upper Mt Simon A interval. The CO_2 saturation increased from a maximum of 30% in 2018 to 44% and the average from 19% to 26%. The total injection volume was 1,094,643 tonnes at the time of the March 2019 PNX survey.

The May 2018 and March 2019 PNX interpretations indicate small amounts of CO_2 in the Mt Simon Upper A and lower portion of Mt Simon B formations up to 6,650 ft., above the 6,840 to 6,830 ft. zone discussed in the previous paragraph. This interval of the Mt. Simon A and Mt Simon B has some fresher fluid remaining in the near wellbore from the 2011 well completion. During well construction, drilling mud filtrate which was much lower in salinity than the formation water invaded the near wellbore formation. Sigma has a depth of investigation of around 10 to 12 inches, so in intervals with drilling mud invasion the measurement will contain both formation and drilling mud filtrate waters. Fresher fluids remaining in the near wellbore is determined from the response of Sigma, TPHI and FNXS. TPHI and FNXS measurements have very low sensitivity to the salinity compared to CO_2, and they indicate the formation fluid is water. Sigma is too low for this water to be native formation water salinity. Also, Sigma is continuing to increase as the near wellbore invaded water diffuses and is replaced by native formation water. This Sigma increase is most evident in the interval 6,780 to 6,815 ft. where most of the non-native near wellbore fluid has diffused. TPHI and FNXS measurements show almost no change indicating only a salinity change has occurred causing Sigma to increase. The interpretation challenge with Sigma increasing from a salinity change is detecting a small amount of CO_2 which may only suppress the increase. In this case, formation CO_2 saturation is largely dependent on the TPHI and FNXS measurements. An additional complication is the CO_2 filled borehole. For a CO_2 filled borehole, TPHI and FNXS have large corrections which adds some uncertainty. In the interval 6,830 to 6,650 ft., the small amount of formation CO_2 calculated is close to the uncertainty of TPHI and

FNXS measurements in a large CO_2 borehole correction. The CO_2 calculated above 6,830 to 6,650 ft. is within the uncertainty of the measurements and below the threshold of definite detection.

Previous fluid interpretations prior to Dec 2014 using RST data indicated small amounts of CO_2 in the upper Mt. Simon A which is believed to be erroneous, particularly for the depth interval 6,707 to 6,813 ft. This CO_2 is not present in interpretations using the newer PNX log data. The RST Pulsed Neutron Log interpretations used Sigma PHIC neutron porosity which has much more uncertainty than the self-correcting algorithms of the PNX. The PNX's neutron porosity, TPHI, has better precision and accuracy and does not give the erroneously interpreted CO_2, while maintaining sensitivity to saturation changes. In addition to a more precise and accurate neutron porosity compared to the RST, PNX has the new FNXS measurement which is sensitive to CO_2 and very little sensitivity to water salinity. FNXS varies linearly with CO_2 volume and can be used as an additional input to solve for CO_2 saturation. FNXS was used for the July 2014, October 2016, May 2018 and March 2019 PNX logs to evaluate the formation fluids.

Most of drilling mud filtrate in the lower Mt. Simon A is believed to be diffused or displaced by the CO_2 injection leaving only formation water and CO_2 as possible pore fluids. Any CO_2 present would decrease Sigma and the neutron porosity with both measurements reinforcing each other in the interpretation. As a result, the RST CO_2 saturation in the lower Mt. Simon A has a higher confidence than zones that have diffusing drilling mud filtrate. The confidence in the PNX interpretations using FNXS does not change for the zones with and without diffusing drilling mud filtrate.

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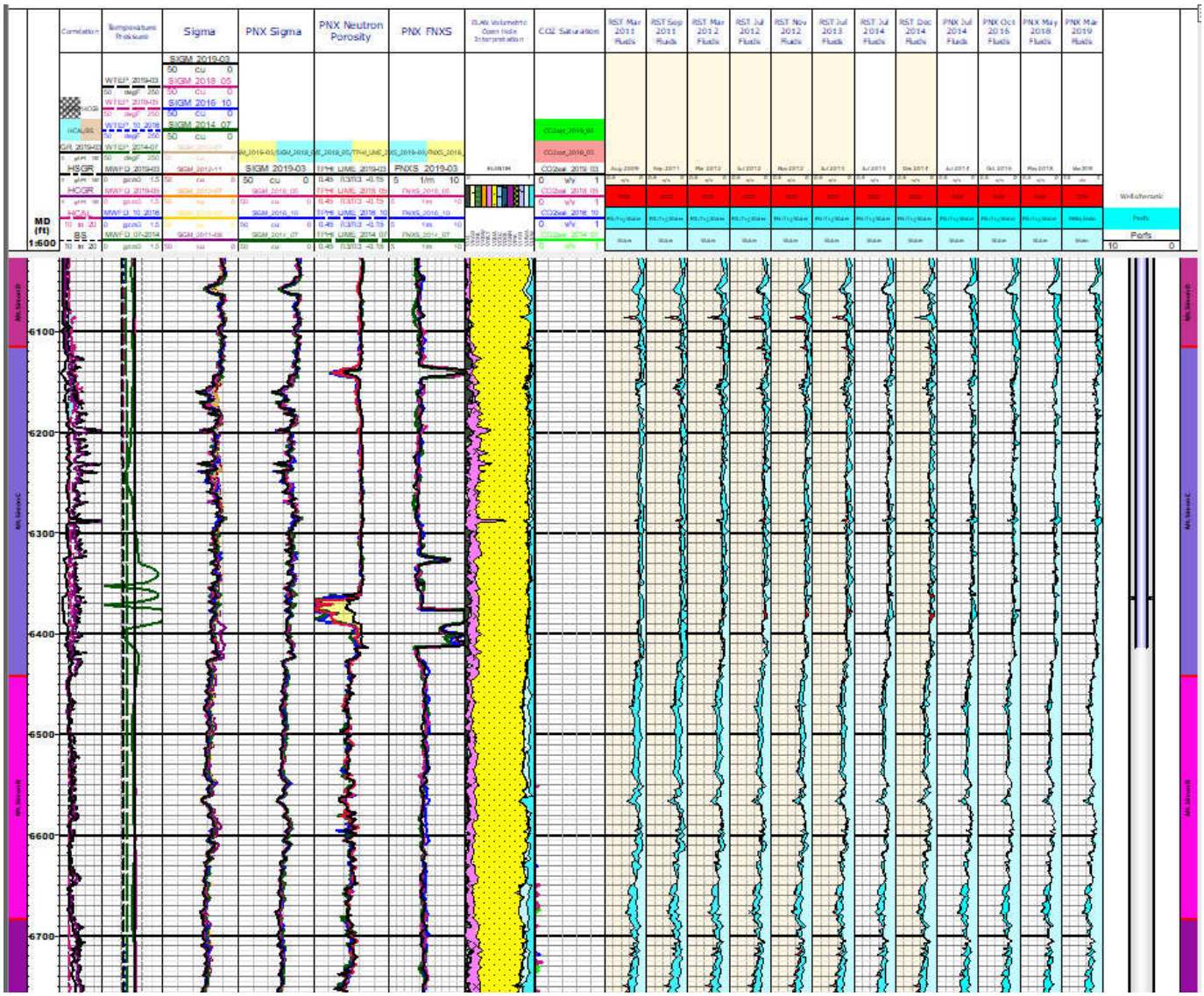


Figure 5: RST and PNX time-lapse fluids interpretation for the depth range 6,030 to 6,750 ft.

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3. Conclusions

A Pulsed Neutron eXtreme (PNX) log was run to evaluate well integrity and formation CO_2 saturation in the CCS1 well on March 6, 2018. Measurements sensitive to annular CO_2 were inspected and compared to previous well integrity logs. The Illinois Basin – Decatur Project injected approximately one million tonnes of CO_2 through perforations in the CCS1 well into the lower Mt. Simon A over a three-year period from November 17, 2011 to November 26, 2014. Pulsed Neutron Log (PNL) monitoring during the injection period showed the majority of the CO_2 to be contained in the lower Mt Simon A interval 6,940 – 7,050 ft. with a small amount extending up into the bottom of the Upper Mt Simon A to 6,912 ft. CO_2 saturations remained fairly constant throughout the injection period with CO_2 segmented into the permeable formation layers. The average CO_2 saturation in the interval 6,940 – 7,050 ft. averaged 29% with a maximum saturation of 56%.. Post injection PNL monitoring in October 2016 showed very little change in CO_2 saturations apart from some buoyancy effects reducing CO_2 saturations at the bottom of the permeable zones, and the average CO_2 saturations remaining constant.

The May 2018 PNX monitoring log showed a significant reduction of the CO_2 saturation with average CO_2 saturation in the interval 6,940 – 7,050 ft. dropping from 29% to11% and maximum from 56% to 31%. This CO_2 saturation level is closer to PNL measured saturations at the VW1 observation well during the injection period. This may indicate high CO_2 saturations concentrated near the injection well may have remained until CO_2 saturations further from the well had time to diffuse to a more uniform level in the Lower Mt Simon. The March 2019 log interpretation showed similar CO_2 saturations near the injection interval although with some increase in height. The May 2018 interpretation shows the majority of the CO_2 in the interval 6,940 – 7,035 ft and a small amount at 6,920 – 6,924 ft. The March 2019 interpretation shows the lower interval expanded 5 ft. to 6,940 – 7,040 ft and the upper interval expanded 16 ft. to 6,904 – 6,924 ft.

The May 2018 PNX log showed a new interval with CO_2 in the Upper Mt Simon A at 6,840 to 6,830 ft. The IL-ICCS project began CO_2 injection in the Upper Mt Simon A and Mt Simon B formation on April 7, 2017 and had injected 687,440 tonnes at the time of the May 2018 CCS1 log. The CO_2 detected in the May 2018 PNX survey at 6,840 to 6,830 ft. is likely a result of the IL-ICCS CO_2 storage. The March 2019 PNX log shows an increase in CO_2 in this same Upper Mt Simon A interval. The CO_2 saturation increased from a maximum of 30% in 2018 to 44% in 2019 and the average from 19% to 26%. The total injection volume was 1,094,643 tonnes at the time of the March 2019 PNX survey.

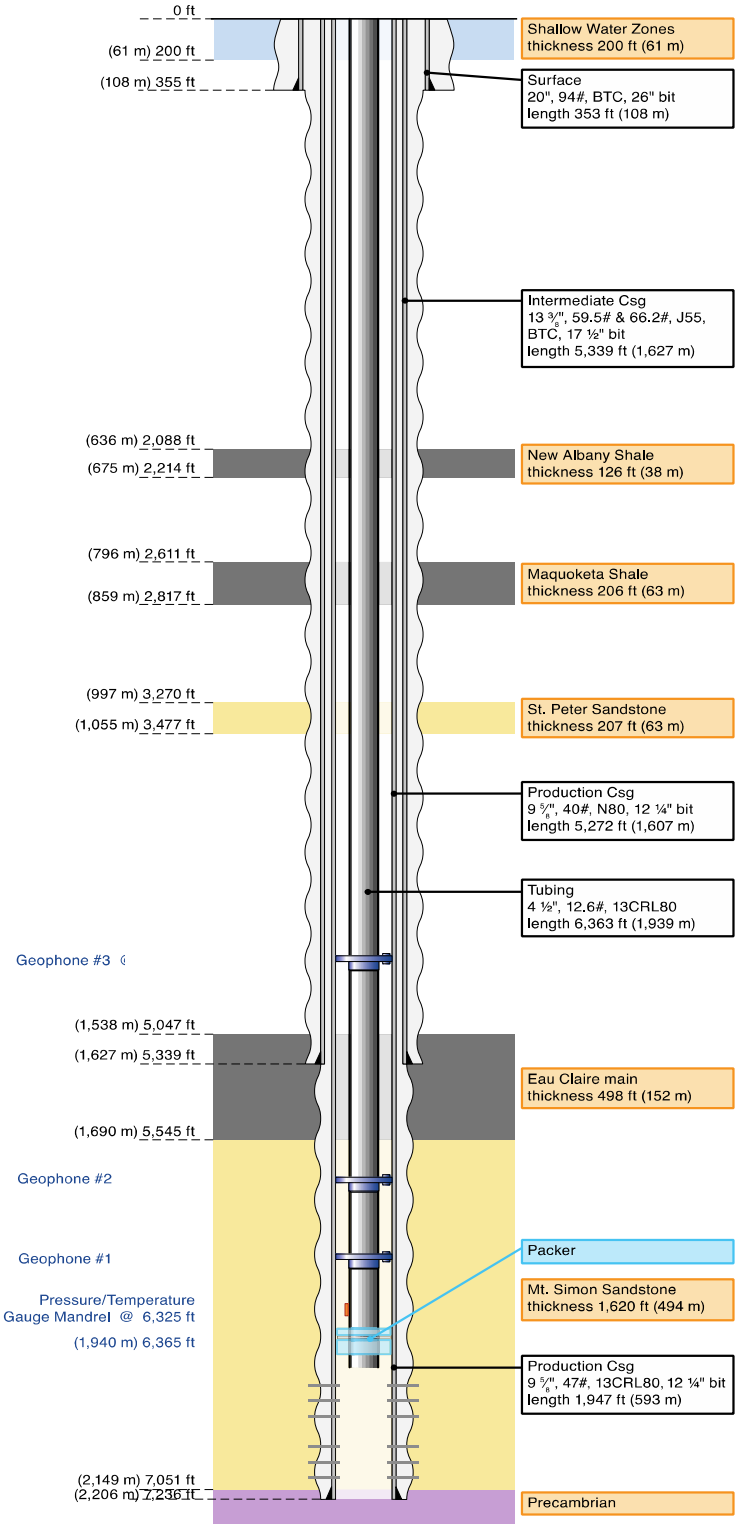
The May 2018 and March 2019 PNX interpretations indicate small amounts of CO_2 in the Mt Simon Upper A and lower portion of Mt Simon B formations up to 6,650 ft., above the 6,840 to 6,830 ft. zone discussed in the previous paragraph. Most of the interval above 6,830 ft to 6,650 ft. still has water in the near wellbore with lower salinity than the formation water. This lower salinity water comes from the well construction and continues to diffuse with time and being replaced with native formation water. The diffusion of the non-native water causes Sigma to increase and countering any decrease from small amounts of CO_2. In this case, formation CO_2 saturation computation is largely dependent on the TPHI and FNXS measurements. An additional complication is the CO_2 filled borehole. For a CO_2 filled borehole, TPHI and FNXS have large corrections which adds some uncertainty. The CO_2 calculated above 6,830 to 6,650 ft. is within the uncertainty of the measurements and below the threshold of definite detection.

The March 2019 PNX survey indicates no change to annular fluids in either the casing to formation or tubing to casing annuli were detected above 6,650 ft. The PNX mechanical integrity log indicates CO_2 is contained within the tubing and casing of the borehole and in areas of the reservoir where expected. There is nothing in the data to indicate that the mechanical integrity of the well has been compromised.

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4. Appendix – CCS1 Completion



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5. References

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